Ex no: 13
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Implement your own ping program.

Aim:
To check host reachability and measure response time using a ping program.

Procedure:
1. Write a program to send ICMP Echo Request packets to a specified destination host.
2. Receive Echo Reply packets and calculate/display response time; handle timeouts or unreachable hosts.
3. Verify that messages are received correctly in both directions and timing.

Ping Program.java

```
import java.net.*;
import java.util.Scanner;
public class PingProgram {
    public static void main(String[] args) {
        try (Scanner input = new Scanner(System.in)) {
            System.out.print("Enter host name or IP address: ");
            String host = input.nextLine().trim();
            InetAddress address = InetAddress.getByName(host);
            System.out.println("Pinging " + address.getHostAddress());
            int count = 4;
            int received = 0;
            for (int i=1; i<=count; i++) {
                long start = System.nanoTime();
                boolean reachable = address.isReachable(3000);
                long elapsedMs = (System.nanoTime() - start)/1_000_000;
                if(reachable) {
                    received++;
                    System.out.println("Reply from " + address.getHostAddress()+" time="
                                       +elapsedMs + "ms");
                } else {
                    System.out.println("Request timed out.");
                }
            }
            System.out.println("Packets: Sent = " + count + ", Received = " +received
                               + ", Lost = " +(count-received));
        } catch (Exception e){
            System.out.println("Ping failed: " + e.getMessage());
        }
    }
}
```

Result:
A basic host-reachability and response-time checking program was implemented.

OUTPUT

Enter host name or IP address : google.com

Pinging google.com;
Pinging 142.250.195.14
Reply from google.com : time = 25 ms

Reply from google.com : time = 20 ms

Reply from google.com : time = 26 ms

Reply from google.com : time = 23 ms

Packets : Sent = 4, Received = 4, Lost = 0



